



Bayesian neural networks for genomic prediction: uncertainty quantification and SNP interpretation with SHAP and GWAS

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Received: 3 June 2025 / Accepted: 14 December 2025

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Abstract

Key message This study presents a Bayesian neural networks framework with LASSO regularization and the GSMESP interpretability tool, enabling accurate, uncertainty-aware, and biologically interpretable genomic prediction.

Abstract Deep learning offers significant potential for genomic prediction by modeling complex, nonlinear genotype–phenotype relationships. However, its application in plant breeding has been constrained by limited model interpretability and a lack of uncertainty quantification. To address these challenges, we developed a Bayesian neural networks (BNNs) framework incorporating least absolute shrinkage and selection operator (LASSO) regularization for multi-trait genomic prediction with credible uncertainty estimation. In parallel, we introduce GSMESP, a novel interpretability framework that integrates SHapley Additive exPlanations (SHAP) with genome-wide association study (GWAS) signals to prioritize trait-associated single nucleotide polymorphisms (SNPs) from both statistical and biological perspectives. We applied this framework to a diverse panel of 1385 upland cotton (*Gossypium hirsutum*) accessions genotyped with over 12,000 SNPs, evaluating performance across multiple fiber-related traits. The BNNs model consistently outperformed conventional and deep learning benchmarks, achieving 0.46–47.85% improvements in predictive accuracy. Moreover, it generated trait- and sample-specific 95% credible intervals, enabling robust uncertainty quantification and more informed selection decisions. Using GSMESP, we identified biologically meaningful loci, with a substantial proportion of top-ranked SNPs located in the D-subgenome. Notably, chromosome D05 emerged as a genomic hotspot enriched for SNPs associated with fiber length, lint percentage, and uniformity. By integrating high predictive performance, credible uncertainty estimation, and biologically grounded interpretability, our framework provides a transparent and robust deep learning approach to accelerate genomic selection in crop breeding programs.

Keywords Genomic selection · Bayesian neural networks · Least absolute shrinkage and selection operator · SHapley Additive exPlanations · Genome-wide association study · GSMESP

Abbreviations

GS Genomic Selection
 BNNs Bayesian Neural Networks

LASSO Least Absolute Shrinkage and Selection Operator
 SNP Single-Nucleotide Polymorphisms
 SHAP SHapley Additive exPlanations
 GWAS Genome-Wide Association Study

Communicated by Huihui Li.

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GSMesP	GWAS meta SHAP
LY	Lint Yield
LP	Lint Percentage
LEN	Fiber Length
SFI	Short Fiber Index
EL	Elongation
UNI	Uniformity
STR	Strength
MIC	Micronaire

Introduction

Genomic selection (GS), originally proposed by Meuwissen et al. (2001), has revolutionized the way complex traits are predicted in plant and animal breeding. By utilizing genome-wide molecular markers to estimate genomic estimated breeding values (GEBVs), GS allows breeders to select superior genotypes more rapidly and accurately, thereby reducing breeding cycles and increasing genetic gain. The success of GS relies on statistical and machine learning models capable of capturing the underlying genetic architecture of complex traits.

Over the years, numerous models have been developed and applied to genomic prediction, generally categorized into four major groups: (i) linear mixed models, such as ridge regression best linear unbiased prediction (rrBLUP) and genomic BLUP (GBLUP); (ii) Bayesian methods including BayesA, BayesB, BayesC π , and BayesR; (iii) nonparametric machine learning algorithms such as random forests (RF) and support vector machines (SVMs); and (iv) deep learning (DL) approaches, including multilayer perceptron (MLP) and convolutional neural networks (CNNs) (Crossa et al. 2017; Azodi et al. 2019). These models have contributed substantially to the prediction of quantitative traits across diverse species.

Despite significant progress, effectively implementing GS in practical plant breeding remains challenging. Many agronomic traits are influenced by complex genetic architectures involving polygenicity, epistasis, genotype-by-environment (G×E) interactions, and nonadditive effects. Traditional linear models often fall short in capturing these complexities, thereby limiting their predictive performance in real-world breeding scenarios (Lorenzana and Bernardo 2009; Xu et al. 2014). Deep learning (DL), however, offers a flexible alternative, capable of modeling intricate nonlinear relationships. It has shown promising results in human, animal, and plant genomics (Bellot et al. 2018; Abdollahi-Arpanahi et al. 2020; Montesinos-López et al. 2019). Early studies, such as deep genomic selection (DeepGS), demonstrated the potential of CNNs in modeling genotype–phenotype associations (Ma et al. 2018). As technology advances, the focus has shifted toward integrating more complex biological data.

For instance, DNNGP significantly improved prediction accuracy by incorporating multi-omics data, further demonstrating deep learning’s ability to handle complex biological information (Wang, et al., 2023a).

Recent advances in deep learning for genomic prediction have moved well beyond the initial use of MLP and CNNs architectures. Transformer-based models, residual neural networks (ResNet), and long short-term memory (LSTM) networks are emerging as next-generation frameworks capable of integrating multiple omics and environmental data. For example, Cropformer employs transformer encoders and attention pooling mechanisms to capture long-range marker dependencies and more effectively manage multi-scenario crop data (Wang, et al., 2023b). Hybrid models like LSTM–ResNet have also emerged to mitigate the vanishing gradient problem and extract hierarchical features from genetic markers, yielding superior predictive performance (Li et al. 2025). Additionally, research has shifted from general-purpose models to those designed to address specific challenges in breeding practice, driving innovations like multitask learning frameworks. For instance, MtCro predicts multiple traits simultaneously, which is crucial for breeding programs targeting complex trait combinations (Chao et al. 2025). Specialized, user-friendly tools such as SoyDNGP for soybean (Gao et al. 2023), AutoGP for maize (Wu et al. 2025), and WheatGP for wheat (Wang et al. 2025) demonstrate the feasibility of deploying deep learning pipelines in breeding programs. However, the complexity of DL architectures, while boosting performance, often compromises model interpretability, further complicating the decision-making process.

Thus, despite these advancements, two critical challenges persist in the broader application of deep learning in GS: model interpretability and robust uncertainty quantification. Interpretability is a critical requirement in GS, where breeding decisions must be biologically justified and actionable. While deep learning models often yield high predictive performance, their “black-box” nature, with numerous parameters and nonlinear transformations, obscures the influence of individual input features (Huang et al. 2024). In biological applications, understanding why a model makes a prediction can be just as important as how accurately it predicts (Sapoval et al. 2022). To address this challenge, explainable artificial intelligence (XAI) tools such as SHapley Additive exPlanations (SHAP) have emerged as powerful approaches for interpreting complex models. SHAP enables both global and local interpretation by quantifying the contribution of each input feature to the model’s output (Lundberg and Lee 2017). Recent studies have demonstrated SHAP’s utility in genomics. For example, Johnsen et al. (2021) used SHAP with tree ensemble methods to uncover gene–gene and gene–environment interactions in genome-wide association studies (GWAS), while Rao et al. (2025) applied SHAP to

improve the stability and interpretability of genomic predictions. Despite offering post hoc explanations, integrating SHAP insights with sophisticated deep learning architectures to provide trait-specific, biologically actionable information remains a challenge.

An equally critical aspect of genomic selection is uncertainty quantification (Sapoval et al. 2022). While most GS models generate only point estimates, breeders often need insight into the confidence associated with these predictions, especially when making high-stakes decisions among top-performing genotypes. Bayesian neural networks (BNNs) offer a principled framework for uncertainty quantification by introducing stochastic weights into neural networks, transforming them into uncertainty-aware models (Carvalho et al. 2023). This approach enables the estimation of two distinct types of uncertainty: aleatoric uncertainty, which stems from inherent data noise such as measurement errors or environmental variability, and epistemic uncertainty, which reflects the model's limited knowledge due to insufficient or unrepresentative training data (Kendall and Gal 2017; Carvalho et al. 2023). Incorporating these uncertainty estimates allows breeders to differentiate between high- and low-confidence predictions, supporting more informed, transparent, and risk-aware selection strategies.

To address the dual challenges of interpretability and uncertainty in deep learning-based genomic prediction, we propose a comprehensive framework combining BNNs and SHAP with GWAS signals. Specifically, our objectives are: (1) to develop a LASSO (least absolute shrinkage and selection operator)-regularized BNNs for multi-trait genomic prediction, enabling high predictive accuracy alongside credible uncertainty estimation, (2) to apply SHAP for model interpretability, identifying SNP-level contributions to predictions both globally and locally, and (3) to integrate SHAP-derived feature importance with GWAS-based statistical associations in a novel framework termed GSMSP (GWAS meta SHAP), providing biologically grounded prioritization of trait-associated SNPs.

This study is conducted on a well-characterized panel of 1,385 cotton (*Gossypium hirsutum*) lines, assessed over 25 years and genotyped with over 12,000 SNPs. The framework is applied to eight agronomic traits, offering a comprehensive assessment of its predictive performance, interpretability, and utility in breeding decisions. To assess generalizability, the BNNs were further validated on a wheat (*Triticum aestivum* L.) dataset comprising 634 recombinant inbred lines and 44,826 SNPs. Our findings demonstrate that the proposed BNNs-based approach not only improves prediction accuracy but also supports uncertainty-aware selection and interpretable SNP prioritization. This study provides a transparent, robust, and actionable deep learning framework to accelerate genomic selection in crop improvement programs.

Materials and methods

Dataset description

The cotton dataset used in this study comprised 1,385 upland cotton lines evaluated across 42 field trials conducted over 25 years (1993–2017) under fully irrigated conditions as part of the Commonwealth Scientific and Industrial Research Organisation (CSIRO) cotton breeding program in Australia (Li et al. 2022). Trials were conducted at the Australian Cotton Research Institute (ACRI) under standardized high-input agronomic practices on gray Vertosol soils in a semiarid climate. Phenotypic traits included lint yield (LY), lint percentage (LP), and six fiber quality parameters: fiber length (LEN), short fiber index (SFI), elongation (EL), uniformity (UNI), strength (STR), and micronaire (MIC), all measured using standardized protocols. Some lines were phenotyped in multiple years and trials, and individuals with missing data were excluded. In total, 1,777 phenotypic observations were retained for analysis. Genotyping was performed using the CottonSNP63K array, yielding 12,296 high-quality biallelic SNPs distributed across all 26 chromosomes (A01–A13 and D01–D13) at an average density of six SNPs per Mbp. SNPs were encoded as -1 (AA), 0 (AB), and 1 (BB).

For independent model evaluation, a wheat dataset from Sandhu et al. (2021) was used. This dataset consists of 650 recombinant inbred lines (RILs) derived from 26 nested association mapping (NAM) families, evaluated between 2014 and 2016 at the Spillman Agronomy Farm in Pullman, WA, USA. Five agronomic traits were measured: grain yield (GY), grain protein content (GPC), heading date (HD), plant height (PH), and test weight (TW). Genotyping employed both the Illumina 90K SNP array and genotyping-by-sequencing (GBS), initially producing 73,345 markers. After quality filtering, 634 RILs and 44,826 SNPs were retained for downstream analyses.

Bayesian framework for genomic prediction

In this study, BNNs were employed to model genotype-to-phenotype relationships while simultaneously estimating predictive uncertainty (Jospin et al. 2022). BNNs integrate neural networks with Bayesian inference, allowing for both accurate prediction and robust uncertainty quantification. Unlike conventional neural networks that rely on fixed parameters, BNNs treat model parameters (e.g., weights and biases) as probability distributions (Fig. 1A), thereby capturing uncertainty in predictions (Charnock et al. 2022; Carvalho et al. 2023).

The training objective in the BNNs is to estimate the posterior distribution of the network parameters given the



Fig. 1 Overview of the Bayesian neural networks (BNNs) architecture and the GSMESP framework for genomic prediction and SNP prioritization. **A** Comparison between a conventional neural network (left) and a Bayesian neural network (right). Circles represent input, hidden, and output layers (top to bottom), with each layer color-coded for clarity. **B** Schematic of the BNNs model. Dimensionally reduced SNP markers and corresponding phenotypic traits serve as inputs, passing through the input layer and multiple hidden layers with dropout, before reaching the output layer to generate trait predictions. **C** Workflow of the GSMESP method. Step 1: Extract SHAP-derived feature importance and identify trait-associated SNPs from GWAS. Step 2: Calculate SNP rank values based on SHAP contributions and GWAS effect sizes. Step 3: Integrate the two rankings using a weighted scheme to obtain a composite importance score for each SNP. Step 4: Output trait-specific SNP importance scores as determined by the GSMESP framework

observed dataset. This is expressed as (Magris and Iosifidis 2023):

$$p(\theta|D_x, D_y) = \frac{p(D_y|D_x, \theta)p(\theta)}{\int_{\Theta} p(D_y|D_x, \theta')p(\theta')d\theta'} \quad (1)$$

where $\theta \in \Theta$ denotes the parameter vector and $p(\theta)$ denotes a prior distribution over the parameters. The training dataset D consists of input features D_x in matrix and corresponding target variables D_y in vector, $p(\theta|D_x, D_y)$ is the posterior distribution, and $p(D_y|D_x, \theta)$ is the likelihood term.

Variational inference

Letting $D = \{D_x, D_y\}$ denote the training dataset, the formula (1) can be expressed as follows (Magris and Iosifidis 2023):

$$p(\theta|D) = \frac{p(D, \theta)}{p(D)} = \frac{p(D|\theta)p(\theta)}{p(D)} \quad (2)$$

where $\theta \in \Theta$ denotes the parameter vector, $p(\theta)$ denotes a prior distribution over the parameters, $p(\theta|D)$ is the posterior distribution, $p(D|\theta)$ is the likelihood, $p(D)$ represents the marginal likelihood, and $p(D) = \int_{\Theta} p(D|\theta)p(\theta)d\theta$.

Exact inference in BNNs is computationally intractable. Therefore, we employed Variational Inference (VI) to approximate the true posterior $P(\theta|D)$ with a tractable variational distribution $q(\theta)$. VI reframes Bayesian inference as an optimization problem: Find the best approximation $q^* (\theta) \in Q$, where Q is a family of distributions, by minimizing the Kullback–Leibler (KL) divergence between $q(\theta)$ and the posterior (Magris and Iosifidis 2023):

$$q^* = \arg \min_{q \in Q} KL(q||p(\theta|D)) = \arg \min_{q \in Q} \int q(\theta) \log \frac{q(\theta)}{p(\theta|D)} d\theta \quad (3)$$

According to the definition of KL divergence and eq. (2):

$$KL(q||p(\theta|D)) = - \int q(\theta) \log \frac{p(D|\theta)p(\theta)}{q(\theta)} d\theta + \log p(D) \quad (4)$$

Since $\log p(D)$ is a constant independent of the model parameters, minimizing the KL divergence is equivalent to maximizing the evidence lower bound (ELBO):

$$L(q) := \int q(\theta) \log \frac{p(D|\theta)p(\theta)}{q(\theta)} d\theta = E_q[\log \frac{p(D|\theta)p(\theta)}{q(\theta)}] \quad (5)$$

Uncertainty estimation

The BNNs framework facilitates the quantification of predictive uncertainty, helping to mitigate overconfident decisions in the presence of uncertain predictions. Due to its probabilistic nature, a BNNs can yield different outputs for the same input by sampling from the posterior distribution of its parameters. To capture this uncertainty, we employed a multiple sampling strategy during the testing phase. Specifically, for each forward pass, a unique set of weights was sampled from the posterior distribution, and the model generated a corresponding prediction. This process was repeated 200 times per sample, producing a predictive distribution. From this distribution, the mean, variance, and standard deviation were computed, and 95% credible intervals were derived. These uncertainty metrics enabled robust characterization of prediction reliability at the individual sample level and supported downstream decision-making under uncertainty.

Feature selection and BNNs architecture

Feature selection was conducted using the LASSO regression. LASSO penalizes the absolute value of regression coefficients, effectively shrinking some coefficients to zero and selecting a sparse set of predictive SNPs (Ranstam et al. 2018). The LASSO optimization objective is defined as:

$$\text{Loss} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^p |\beta_j| \quad (6)$$

where y_i represents the target variable, $\hat{y}_i$ is the predicted value, β_j denotes the feature coefficients, α is the regularization parameter controlling the strength of the penalty term, n is the number of samples, and p is the number of SNPs.

After feature extraction via LASSO, SNPs with the most significant contributions to phenotypic traits were identified. The selected SNPs were then used as inputs to the BNNs model. In this study, the BNNs architecture (Fig. 1B) consisted of an input layer corresponding to the number of selected SNPs and corresponding phenotypic traits, three hidden layers, each followed by a dropout layer, and a single-node output layer for trait prediction.

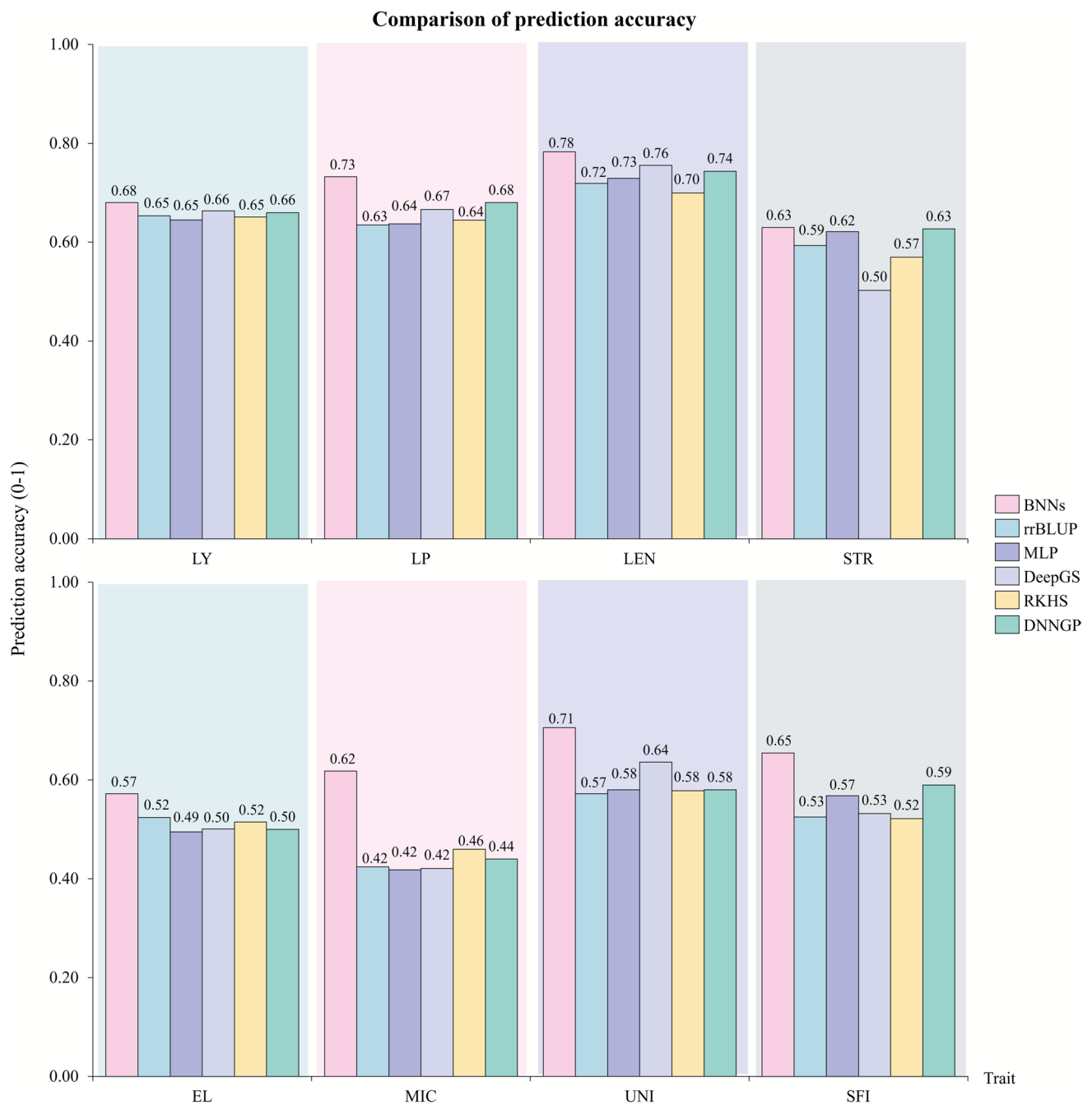


Fig. 2 Comparison of predictive performance across six genomic selection models in cotton. Prediction accuracy (y-axis) was evaluated for eight traits (x-axis) using tenfold cross-validation. Model performance is represented by differently colored bars for the Bayes-

ian neural network (BNNs), ridge regression BLUP (rrBLUP), multi-layer perceptron (MLP), DeepGS, reproducing kernel Hilbert space (RKHS), and DNNGP

Comparative models

To benchmark the BNNs model, five widely used genomic prediction models rrBLUP, MLP, DeepGS, RKHS, and DNNGP were included for comparative analysis. The model configurations and hyperparameters for these baseline methods are described in Supplementary File 1.

Cross-validation and model evaluation

To evaluate the predictive performance and generalization ability of the models, we implemented a tenfold cross-validation (CV) strategy. The full dataset was randomly divided into ten equal subsets. In each iteration, one subset was held out as an independent test set, while the remaining nine

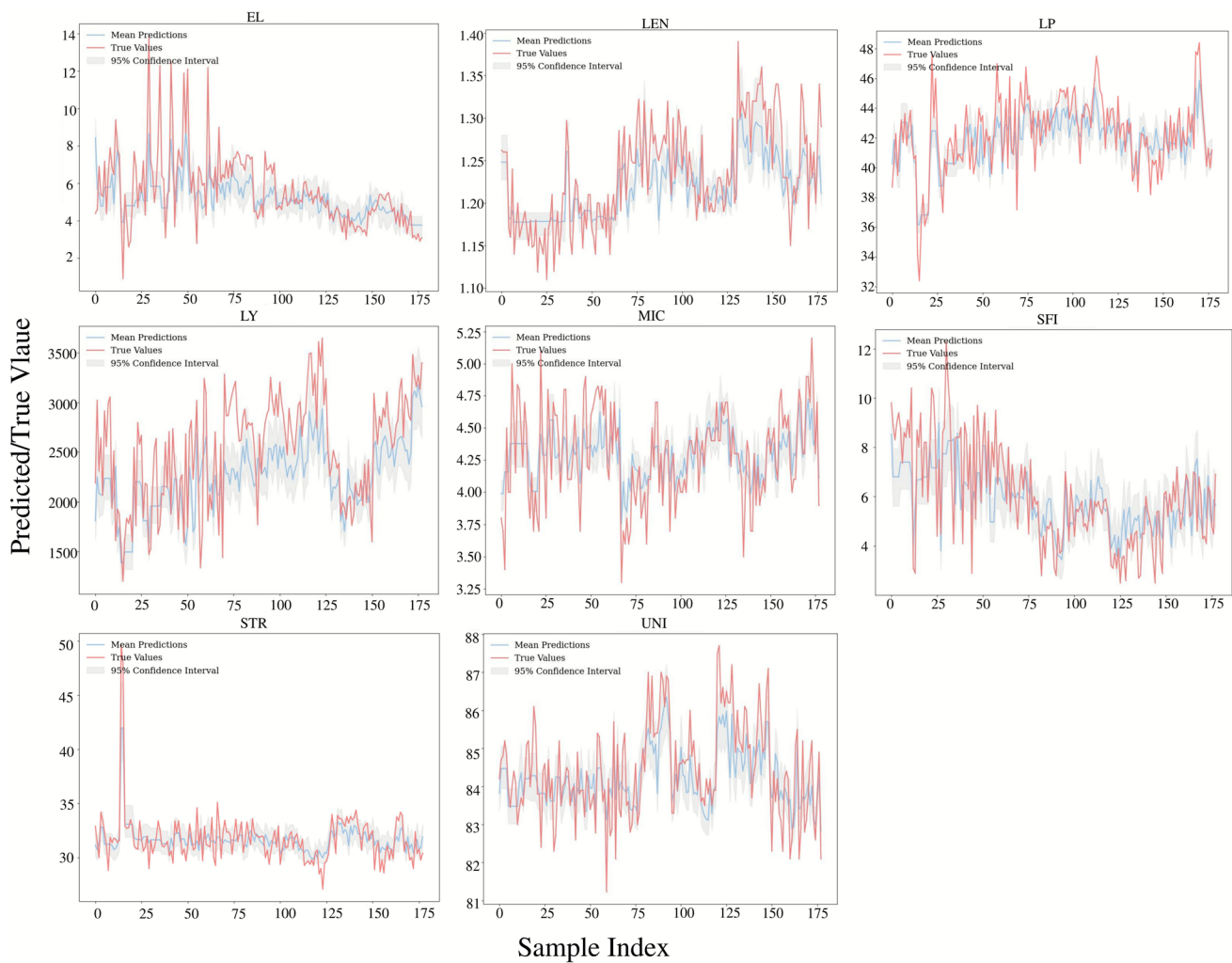


Fig. 3 Uncertainty estimation for test set samples. The x-axis denotes sample indices, while the y-axis shows the values of the trait. The red solid line represents the true value, the blue solid line indicates the

predicted mean from the Bayesian neural networks (BNNs), and the shaded gray area corresponds to the 95% credible interval obtained through repeated sampling

subsets were used for training. This process was repeated ten times so that each subset served once as the test set. The final performance was reported as the average across all folds. Model accuracy was quantified using the Pearson correlation coefficient (PCC) between the predicted and observed phenotypic values in the held-out test set, ensuring an unbiased estimate of predictive performance:

$$\text{PCC} = \frac{\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2}} \quad (7)$$

where $\hat{y}_i$ is the predicted value for the i th sample, y_i is the actual value for the i th sample, $\bar{\hat{y}}$ is the mean of all predicted

values, $\bar{y}$ is the mean of all actual values, and n is the number of samples.

SHAP-based model interpretation

SHAP were used to interpret model predictions by quantifying the contribution of each SNP feature to individual predictions. SHAP values are based on Shapley values from cooperative game theory and provide consistent, local interpretability. The Shapley value for a given feature i is defined as (Lundberg and Lee 2017; Ponce-Bobadilla et al. 2024):

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)] \quad (8)$$

where ϕ_i is the Shapley value for feature i , F is the set of all features, S is a subset of features, and $S \subseteq F$; $S \cup \{i\}$ denotes the union of S and feature i , $f_{S \cup \{i\}}(x_{S \cup \{i\}})$ denotes the model output when trained with subset S and feature i , $f_S(x_S)$ is the model output using only subset S , $f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)$ is the marginal contribution of feature i to subset S , $\frac{|S|!(|F|-|S|-1)!}{|F|!}$ is the weight of the marginal contribution, and $\sum_{S \subseteq N \setminus \{i\}}$ represents sum of all possible features without feature i .

In this study, SHAP analysis was conducted using the Python package SHAP, with the GradientExplainer selected to compute SHAP values for the trained BNNs model. This allowed for the assessment of each feature's contribution to each predicted value. Aggregated SHAP values across all samples were used to rank the importance of SNPs globally, identifying features with the greatest overall influence on model predictions. Furthermore, SHAP enabled local interpretability by evaluating how each SNP contributed to individual predictions. To enhance interpretability and transparency, SHAP visualization tools such as beeswarm summary plots and dependence plots were used to illustrate the model's internal decision-making process and to support biological insights into SNP–trait associations.

Genome-wide association study

A genome-wide association study (GWAS) was conducted using a linear mixed effects model, where genotype data were treated as fixed effects and family structure was modeled as a random effect (Li and Zhu et al. 2013; Hsu et al. 2018). In addition, p-values for marker–trait associations were adjusted using the false discovery rate (FDR) method.

Integrated interpretation via GSMeSP

To combine statistical significance and model-driven insights, we developed GWAS meta SHAP (GSMeSP), a hybrid framework that combines outputs from SHAP and GWAS analyses. SHAP values were averaged across all samples accounting for both positive and negative contributions to rank SNPs according to their overall impact. SNPs were independently ranked based on their SHAP importance scores and GWAS effect sizes. The step-by-step workflow of the GSMeSP method is illustrated in Fig. 1C. A weighted ranking scheme was used to generate a consensus importance score for each SNP:

$$\text{rank}_{\text{GSMeSP}} = \text{rank}(\alpha \text{rank}_{\text{SHAP}} + (1 - \alpha) \text{rank}_{\text{GWAS}}) \quad (9)$$

where α is a weighting coefficient between 0 and 1, reflecting the relative influence of SHAP and GWAS rankings in the final score. In this study, we set $\alpha = 0.6$, giving more

weight to SHAP-derived importance due to its capacity to capture both positively and negatively contributing feature contributions as well as nonlinear interactions.

Hyperparameters setting

To assess model performance, hyperparameters were initially fine-tuned, and the optimal configuration for each model was selected based on the Pearson correlation coefficient and mean squared error (MSE) computed on the validation data. For the BNNs, we imposed a prior distribution on the model weights in the form of a mixture of two zero-mean Gaussian distributions with distinct standard deviations ($\sigma_1 = 1$, $\sigma_2 = 0.00001$), weighted by a mixing parameter ($\pi = 0.5$). Variational inference was employed to approximate the posterior distribution, facilitating efficient Bayesian learning. Input features were selected via LASSO, with the regularization parameter $\alpha = 0.001$ determined through tenfold cross-validation. The architectures of the BNNs comprised three hidden layers for cotton and two hidden layers for wheat, with ReLU activation functions applied in all hidden layers. Model training was conducted using the Adam optimizer with a learning rate of 0.01 for 300 epochs. The ELBO served as the loss function for optimization. Predictive performance was evaluated using tenfold cross-validation to ensure robustness and generalizability. Dropout rates ranging from 0.3 to 0.7 were applied to mitigate overfitting. Hyperparameter configurations for the MLP, DeepGS, RKHS, and DNNGP models are detailed in Supplementary File 1.

Software and package

All experiments and algorithmic models in this study were conducted on a Linux CentOS 7 system equipped with 256 GB of RAM, a 1 TB hard drive, and an NVIDIA RTX 4070 Ti GPU with 16 GB of VRAM. BNNs, MLP, DeepGS, RKHS, and DNNGP models were implemented using Python 3.11, while rrBLUP was executed in R version 4.3.0 using the rrBLUP package (Endelman 2011). SHAP analysis was performed using the Python package SHAP, with the GradientExplainer selected to compute SHAP values for BNNs. GWAS analysis was carried out using the mixedlm function from the Python package statsmodels (version 0.14.4), and corresponding visualizations were produced using the R package CMplot. In the GSMeSP analysis, rank calculations were conducted using the RANK function in Microsoft Excel.

Table 1 Quantitative summary of uncertainty estimates for selected test samples of LP

Sample	True value	Mean prediction	95% CI lower	95% CI upper	Variance
14	34.10	35.5340	31.6955	39.3726	3.8355
28	37.00	37.3100	33.7027	40.9173	3.3873
43	39.90	40.5740	35.1329	46.0152	7.7067
69	37.18	38.1579	33.4582	42.8577	5.7495
79	39.80	40.0776	35.0708	45.0844	6.5255
104	40.30	39.6672	34.6230	44.7114	6.6232
106	43.40	40.7104	34.8674	46.5533	8.8869
109	43.80	39.5763	34.6882	44.4645	6.2198
128	43.00	39.5150	34.4871	44.5428	6.5804
136	38.40	38.2778	34.3023	42.2533	4.1141
152	41.60	39.9563	34.6216	45.2911	7.4082

95% CI Lower, lower bound of the 95% credible interval; 95% CI Upper, upper bound of the 95% credible interval.

Results

Predictive performance of the Bayesian neural networks

We evaluated the phenotypic prediction accuracy of six models: BNNs, rrBLUP, MLP, DeepGS, RKHS, and DNNGP regression. Among these, BNNs consistently outperformed all other models across the eight target traits (Fig. 2) in cotton dataset, demonstrating superior capacity to capture complex genetic architectures. The gain in predictive accuracy ranged from 0.46 to 47.85%, depending on the trait. For LY, all models performed comparably, with BNNs offering a modest gain of 2.50 to 5.44%. For STR, MLP and DNNGP showed similar prediction accuracy as BNNs. In contrast, for MIC, BNNs achieved the most substantial accuracy gain, outperforming competing models by 34.35 to 47.85%. Performance differences across models were also trait-specific: While rrBLUP and MLP showed similar results for most traits, MLP and DeepGS diverged notably for STR, indicating that the models differ in their ability to capture trait-specific genetic effects. Since all models used the same $-1/0/1$ genotype encoding and CNN-based architectures such as DeepGS, DNNGP, and BNNs are input-agnostic, these differences are unlikely to be attributable to encoding schemes. The superior performance of BNNs was further confirmed in an independent wheat dataset (Fig. S1). Across five traits and three breeding cycles, BNNs consistently outperformed all benchmark models, achieving average accuracy gains of up to 34.7% depending on trait and cycle, underscoring the robustness and generalizability of our framework.

Uncertainty estimation from Bayesian predictions

Unlike traditional models that produce single-point estimates, the BNNs framework provides probabilistic predictions, allowing estimation of credible intervals (CIs) and predictive variances. For each sample–trait combination, 95% CIs were calculated (Fig. 3), alongside predicted and observed values. Across most traits, the CIs were relatively narrow and closely aligned with observed phenotypes, indicating high model confidence and prediction reliability (Fig. 3). To provide a quantitative benchmark, we further expressed CI width as a percentage of the phenotypic range for each trait (Table S1). For 6 out of 8 traits, the average CI width represented only 12.3–17.0% of the trait range, underscoring consistently precise predictions. In contrast, traits SFI (35.5%) and LP (62.6%) displayed wider intervals, highlighting them as more challenging prediction targets. Importantly, CI widths were generally stable across samples, reflecting robust model performance throughout the dataset. Nevertheless, individual deviations were observed, as shown in Table 1 for trait LP. For instance, sample 104 had an absolute deviation of 0.6328, indicating moderate predictive error. Moreover, variation in CI widths reflected differences in model certainty: Sample 106 exhibited a wider CI ([34.8674, 46.5533]; range = 11.69), whereas sample 109 had a narrower CI ([34.6882, 44.4645]; range = 9.78), suggesting greater model confidence in the latter prediction.

Model interpretability through SHAP analysis

To provide interpretable SNP-level insights into the predictions of the BNNs models, we computed SHAP values for genomic features preselected via LASSO (Supplementary File 2). Within this framework, a SNP's SHAP value quantifies its marginal contribution to the deviation of an individual sample's prediction from the model's average output

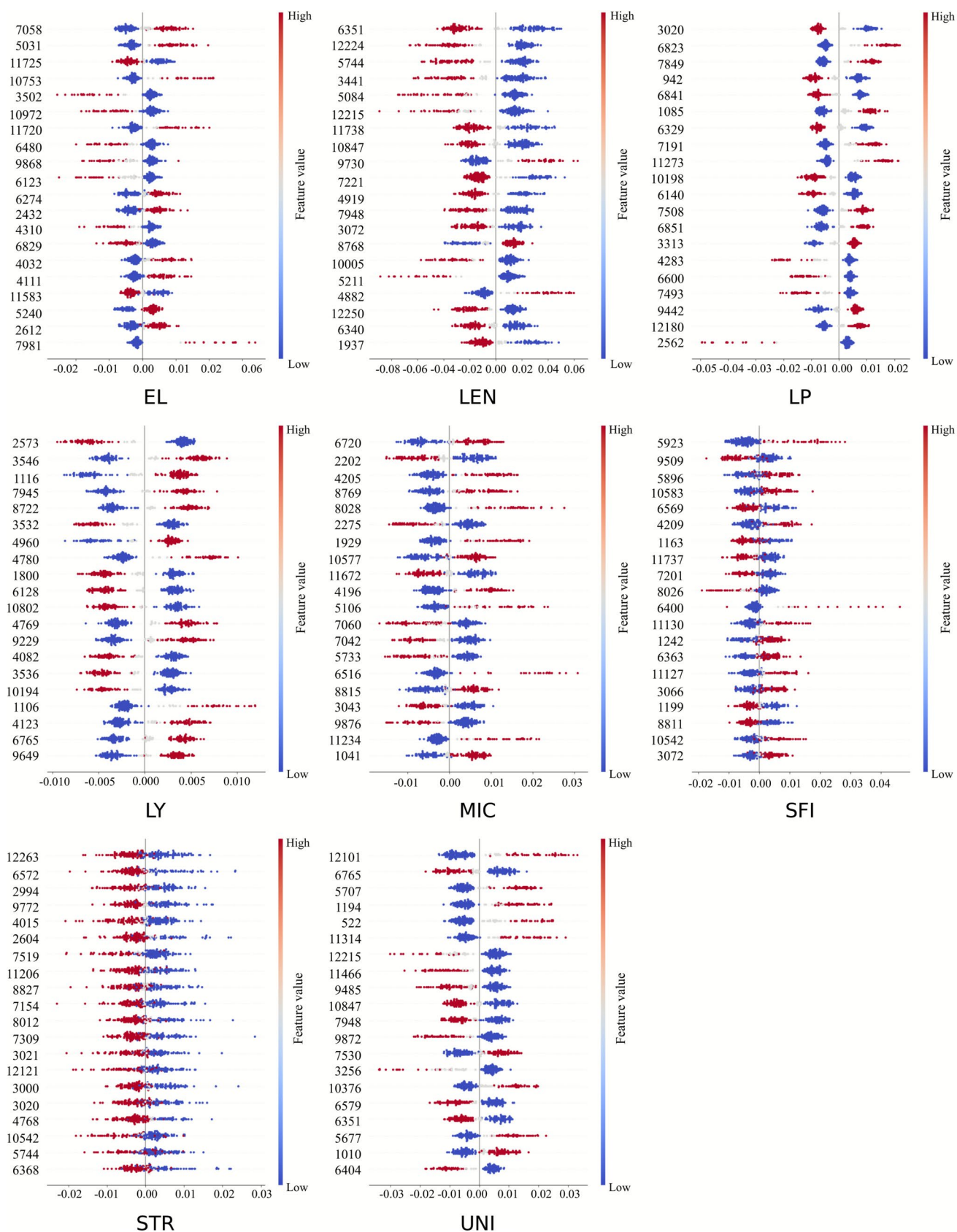


Fig. 4 SHAP beeswarm summary plots for eight cotton traits. Each plot displays the top 20 SNPs ranked by their average contribution to the model's predictions. The x-axis shows the magnitude of each SNP's impact on predicted trait values. The left y-axis lists SNPs in descending order of importance, while the color gradient (blue to red) indicates the feature value of each SNP (blue = -1, red = 1). Each point represents an individual data instance, with its position reflecting the SHAP value and its color indicating the corresponding SNP genotype

(base value). Thus, the final prediction for a sample is the sum of the base value and the SHAP contributions of all SNPs, with positive values increasing and negative values decreasing the predicted trait outcome.

Beeswarm summary plots visualized the top 20 SNPs per trait ranked by their mean absolute SHAP contribution, simultaneously revealing effect direction and distribution (Fig. 4). For LEN, 17 of the top 20 SNPs consistently exhibited positive SHAP values when encoded as -1, indicating that this allele was associated with increased predicted fiber length. The SHAP distributions further reflected fundamental differences in trait architecture. For STR, the diffuse spread of SHAP values indicated that the effects of top SNPs were highly context-dependent and unpredictable, likely modulated by epistasis or environmental interactions. In contrast, for LP, the sharp concentration of similarly sized SHAP values indicated that the effects of its top SNPs were highly stable and reliable across the population, making their contributions to the phenotype more predictable.

To assess overlap in predictive importance across traits, we compared the top 20 SHAP-ranked SNPs among all eight traits. Across 28 pairwise comparisons, shared SNPs were observed in six traits (Table 2). For example, SNP 3020 contributed substantially to both LP and STR, suggesting pleiotropic regulation. Conversely, unique top-ranked SNPs likely represent trait-specific genetic determinants.

Nonlinear effects and SNP-SNP interactions were further examined using SHAP dependence plots for the four most influential SNPs per trait ranked by their mean absolute SHAP contribution (Fig. 5). These plots show the relationship between a SNP's encoded value and its SHAP contribution, with color gradients indicating interaction with a secondary SNP. In EL, SNP 7058 exhibited genotype-dependent reversal of effect: Encoding -1 was associated with negative SHAP values, whereas encoding 1 yielded positive contribution. For LEN, we identified a conditionally modulating interaction between SNPs 6351 and 12224, in which SNP 12224 consistently altered the effect magnitude of SNP 6351. When SNP 6351 was encoded as -1 (positively influencing LEN), the presence of SNP 12224 with genotype -1 exerted a synergistic effect, increasing the SHAP value from approximately 0.01 to 0.05, thereby substantially enhancing predicted fiber length. Conversely, when SNP 6351 was encoded as 1 (negatively influencing

LEN), SNP 12224 partially buffered this negative impact: The SHAP value shifted about -0.045 toward -0.01 as the genotype of SNP 12224 changed from -1 to 1, although the negative effect was not fully eliminated. Thus, SNP 12224 not only exerted a strong individual effect on LEN prediction but also acted as a conditional modifier of SNP 6351. Notably, SNP 12224 was annotated to Gh_D13G2343, a cotton gene homologous to *Arabidopsis* SOL1 (AT3G22760), suggesting a plausible biological mechanism in fiber development. Together, these findings underscore the role of conditional SNP interactions in shaping phenotypic predictions and reveal an asymmetric epistatic relationship that refines our understanding of the genetic architecture of cotton fiber traits.

GSMESP analysis

To complement the SHAP analysis with biological validation, GWAS was conducted for all traits. Manhattan plots (Fig. 6) revealed diverse significance patterns across traits, reflecting varying genetic architectures. For instance, SNPs associated with STR were more broadly distributed, consistent with its diffuse SHAP profile (Fig. 4) and suggesting polygenic control. To prioritize trait-associated SNPs, we developed the GSMESP framework, which integrates SHAP-derived importance with GWAS significance rankings. A composite score was computed using a weighted scheme with 60% for SHAP and 40% for GWAS (Table 3; Supplementary File 2). To evaluate sensitivity to the weighting parameter α , we tested values from 0.4 to 0.7. The list of top-ranked SNPs remained highly stable within $\alpha = 0.55 - 0.65$. While relative rankings showed minor shifts, the SNP identities within the Top 50 and Top 200 lists were largely preserved. For example, at least 94% of the Top 50 SNPs and 95% of the Top 200 SNPs identified with $\alpha = 0.6$ remained in the top ranks when α was adjusted to 0.55 or 0.65 (Table S2). This demonstrates that GSMESP's prioritizations, and their biological interpretations, are robust to modest parameter changes. Accordingly, $\alpha = 0.6$ was selected as a representative and balanced value.

For LP, the top SNPs included 6552, 6914, and 10681, while LEN was most strongly associated with SNPs such as 4183, 4222, and 3944. GSMESP also enabled cross-trait comparisons. The top 20 SNPs identified for each trait were compared across traits, revealing five SNPs shared among multiple traits (Fig. 7). For instance, SNP 11159 was highly ranked for both EL and SFI, and SNP 8590 was prioritized for both MIC and SFI. SNP 8590 is located 872 bp downstream of GhCesA4-1-D07 (GH_D07G2083), a known fiber synthesis gene, representing a strong candidate for functional validation. The remaining shared SNPs were located at greater distances from annotated genes, potentially representing previously uncharacterized loci. These

Table 2 Shared SNPs among traits based on the top 20 SHAP-ranked SNPs per trait

Trait 1	Trait 2	Shared SNP ID
LY	UNI	6765
LP	STR	3020
SFI	STR	10542
SFI	LEN	3072
STR	LEN	5744
UNI	LEN	10847; 7948

results demonstrate that GSMeSP identifies loci that are both predictive and biologically meaningful, providing promising targets for future trait improvement studies.

We further investigated the chromosomal distribution of the top 100 SNPs per trait, as prioritized by GSMeSP. The “chromosome carrying rate,” defined as the proportion of top-ranked SNPs located on each chromosome, was computed and visualized to assess trait-specific locus clustering (Fig. 8). GSMeSP effectively highlighted biologically meaningful loci, with a pronounced concentration of significant SNPs mapped to the D-subgenome. Among these,

chromosome D05 exhibited the highest carrying rate for LEN and also contributed substantially to LP and UNI, indicating the presence of key regulatory regions involved in fiber development. Chromosomal contributions varied across traits, reflecting differences in genetic architecture. For example, UNI appeared to be influenced by a limited number of loci, primarily on D05, whereas LP was associated with a broader set of D-subgenome chromosomes, including D03, D05, D07, D10, and D12. Additionally, chromosome A10 showed a strong association with LY, underscoring trait-specific genomic distributions.

Discussion

Enhanced genomic prediction through LASSO-based BNNs

Recent advances in deep learning have significantly enhanced genomic prediction by enabling the modeling of complex, nonlinear interactions among genomic features.

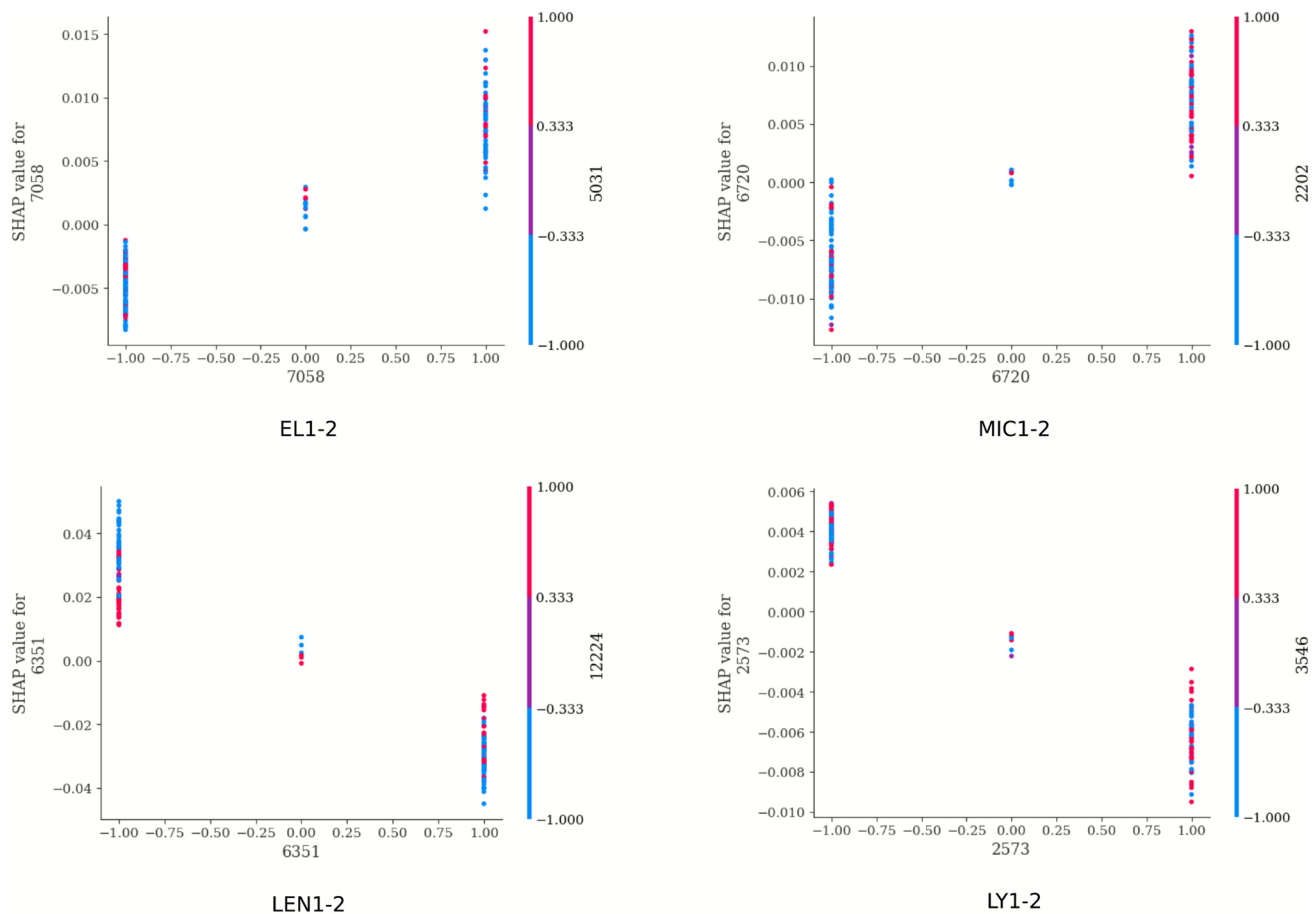


Fig. 5 Dependence plots illustrating SNP interactions. Panels EL1–2, MIC1–2, LEN1–2, and LY1–2 show dependence plots for selected SNPs associated with traits EL (SNPs 7058, 5031), MIC (SNPs 6720, 2202), LEN (SNPs 6351, 12224), and LY (SNPs 2573, 3546), respec-

tively. The x-axis represents the genotype of the primary SNP, the left y-axis shows its SHAP value, and the right y-axis indicates the genotype of the interacting SNP. Each point represents a sample, with color indicating the interacting SNP genotype

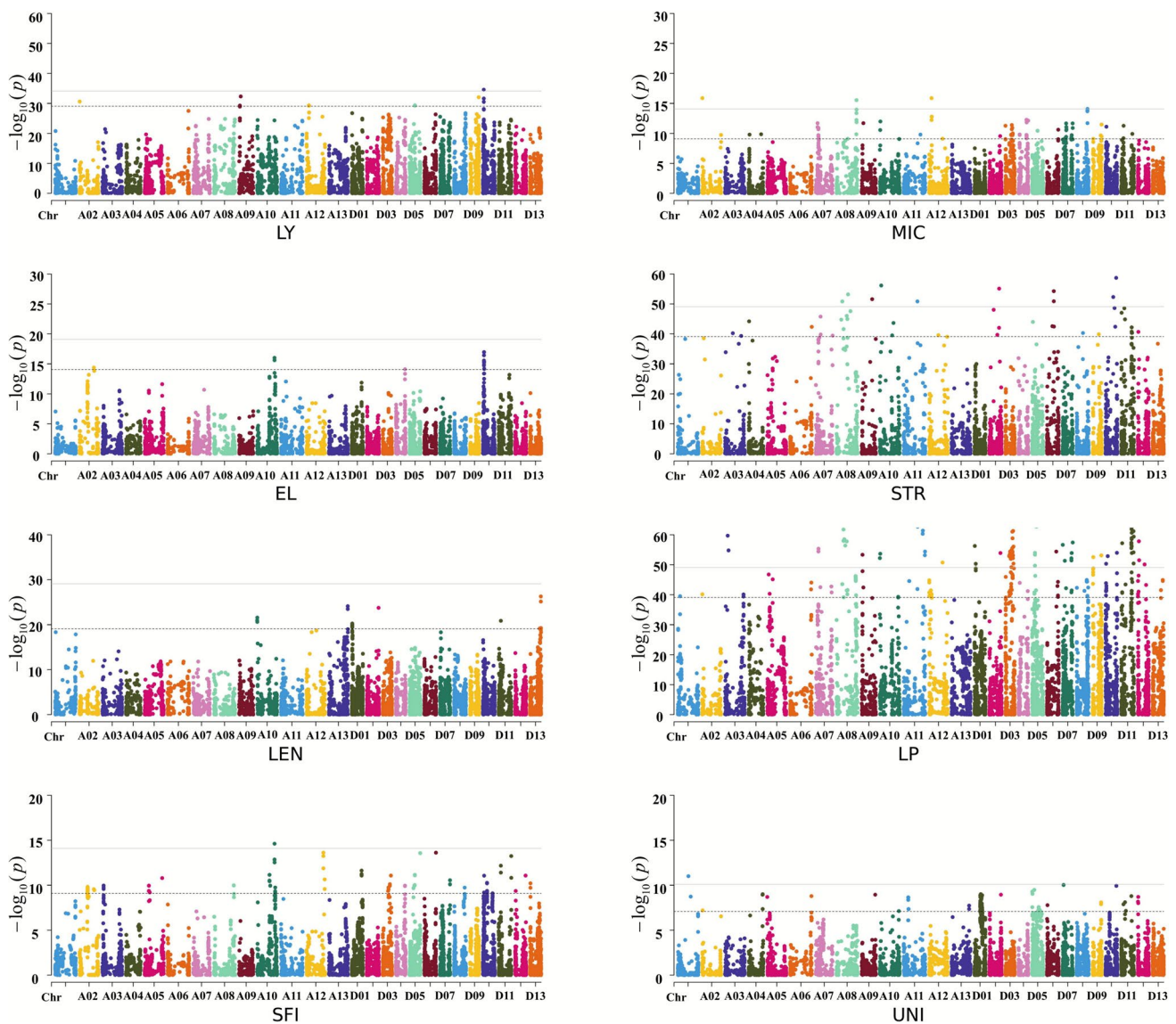


Fig. 6 Manhattan plots for trait association analysis from GWAS. The x-axis represents chromosomes, from left to right, with alternating colors distinguishing different chromosomes. The y-axis shows the statistical significance of each SNP, expressed as $-\log_{10}(\text{P-value})$. Each point represents a SNP, with its x-coordinate corresponding to

its chromosomal position and its height indicating significance. Horizontal solid and dashed lines indicate two significance thresholds, allowing for the visual identification of SNPs significantly associated with the trait

Its growing application in crop breeding reflects its potential to improve prediction accuracy and support data-driven selection strategies (Montesinos-López et al. 2024). In this study, we introduced a LASSO-based BNNs for multi-trait genomic prediction, integrating the variable selection capability of LASSO with the uncertainty quantification strength of Bayesian inference. Across two diverse datasets, this approach consistently outperformed conventional and deep learning methods, including rrBLUP, MLP, RKHS, DeepGS, and DNNGP, achieving an average accuracy gain of 25.4% across multiple agronomic traits and growing cycles (Fig. 2; Fig. S1).

The observed improvement can be attributed to the synergistic effect of LASSO-driven dimensionality reduction and the probabilistic learning framework of BNNs. The common “large p, small n” challenge in genomic prediction, where the number of markers (p) greatly exceeds the number of samples (n), often leads to multi-collinearity and overfitting. LASSO performs simultaneous variable selection and regularization, producing a sparse set of informative markers that enhances interpretability, computational efficiency, and predictive stability (Theng & Bhoyar 2024). While methods such as Elastic Net (EN) may be advantageous under extreme collinearity, LASSO proved both effective

Table 3 Partial SNP ranking based on GSMeSP analysis for eight traits

Trait	SNP	Chromosome	Position	Mean SHAP value	$rank_{SHAP}$	GWAS P_value	$rank_{GWAS}$	$rank_{GSMeSP}$
LY	9673	D08	65342622	0.000379259	11	1.15342E-25	48	1
	2338	A09	9805810	0.000112668	275	5.09147E-33	2	8
LP	6552	D04	53044377	0.000223531	88	2.90178E-39	301	1
	10681	D10	67215776	0.000196841	111	1.8374E-32	571	3
EL	815	A03	97828763	0.000309748	28	3.7677E-11	87	1
	10884	D11	12282892	0.000563798	5	1.59213E-08	181	2
LEN	4183	D01	2318179	5.41E-05	59	1.70215E-20	15	1
	4222	D01	3789622	0.00031577	42	1.23122E-17	46	2
MIC	6059	D03	36499091	3.25E-05	232	4.00097E-12	33	3
	10898	D11	13202473	-0.000333763	541	5.32339E-12	34	14
SFI	2797	A10	70692085	0.000253879	53	7.19757E-12	21	1
	11159	D11	72144636	0.000214872	73	5.79969E-14	7	2
STR	10976	D11	22557978	0.000101729	50	2.60744E-25	241	1
	2034	A08	71159037	0.000144446	10	1.9149E-22	307	2
UNI	4666	D01	44009115	0.000378917	81	3.24018E-08	108	1
	4050	A13	99766572	0.000149431	244	4.39741E-08	118	7

and efficient in this study, consistent with prior evidence in plant and animal breeding (Li et al. 2015; Mathew et al. 2022). Concurrently, BNNs mitigate overfitting and provide robust modeling of imbalanced SNP data, including rare variant effects, by accounting for uncertainty in model parameters (Jospin et al. 2022; Wang et al. 2022). Together, these strengths position the LASSO-based BNNs as a promising and interpretable tool for advancing multi-trait genomic prediction in plant breeding programs.

Quantifying predictive uncertainty for robust selection

An often-overlooked dimension of genomic prediction is the uncertainty associated with predicted values. Although most breeding decisions are based on point estimates of genetic merit, neglecting predictive uncertainty can result in unstable genetic gains and inefficient allocation of resources, particularly when predictions are applied across diverse or fluctuating environments (Villar-Hernández et al. 2018). The ability of BNNs to generate full predictive distributions, rather than fixed point estimates, marks a significant advancement in this context (Ferrão et al. 2017; Jospin et al. 2022). This capability is especially important given that deep learning models are often prone to overconfident predictions (Thuy and Benoit 2024).

In our study, the BNNs framework enabled the estimation of confidence intervals for each genotype–trait combination, providing a probabilistic measure of prediction reliability (Fig. 3). Most traits exhibited relatively narrow, stable intervals, approximately 12–17% of the phenotypic range, highlighting model precision and robustness. In contrast, LP

and SFI showed wider intervals, reflecting more complex genetic architectures or stronger genotype-by-environment interactions. This differentiation underscores how uncertainty estimates can identify traits with reliable predictions versus those requiring additional validation. At the individual level, genotypes with both high predicted values and narrow intervals represent strong candidates for selection, as they are more likely to maintain stable performance across environments. In contrast, wider intervals may signal greater sensitivity to environmental variation, data limitations, or weaker genetic signals, suggesting the need for additional evaluation or supporting data. This capacity is particularly valuable for traits influenced by complex genetic architectures or strong genotype-by-environment interactions. By integrating uncertainty into selection strategies, breeders can prioritize genotypes not only by expected performance but also by the confidence in those predictions. As such, predictive uncertainty becomes a critical decision support tool, complementing traditional selection indices and enhancing the overall reliability of breeding outcomes.

Interpretable modeling through GSMeSP based on SHAP and GWAS analysis

Beyond managing predictive uncertainty, model interpretability remains a major barrier to the practical adoption of deep learning in plant breeding (Sapoval et al. 2022). To address this limitation, we developed GSMeSP, an interpretability framework that integrates SHAP with GWAS, allowing transparent and biologically grounded inference of genotype–phenotype relationships.

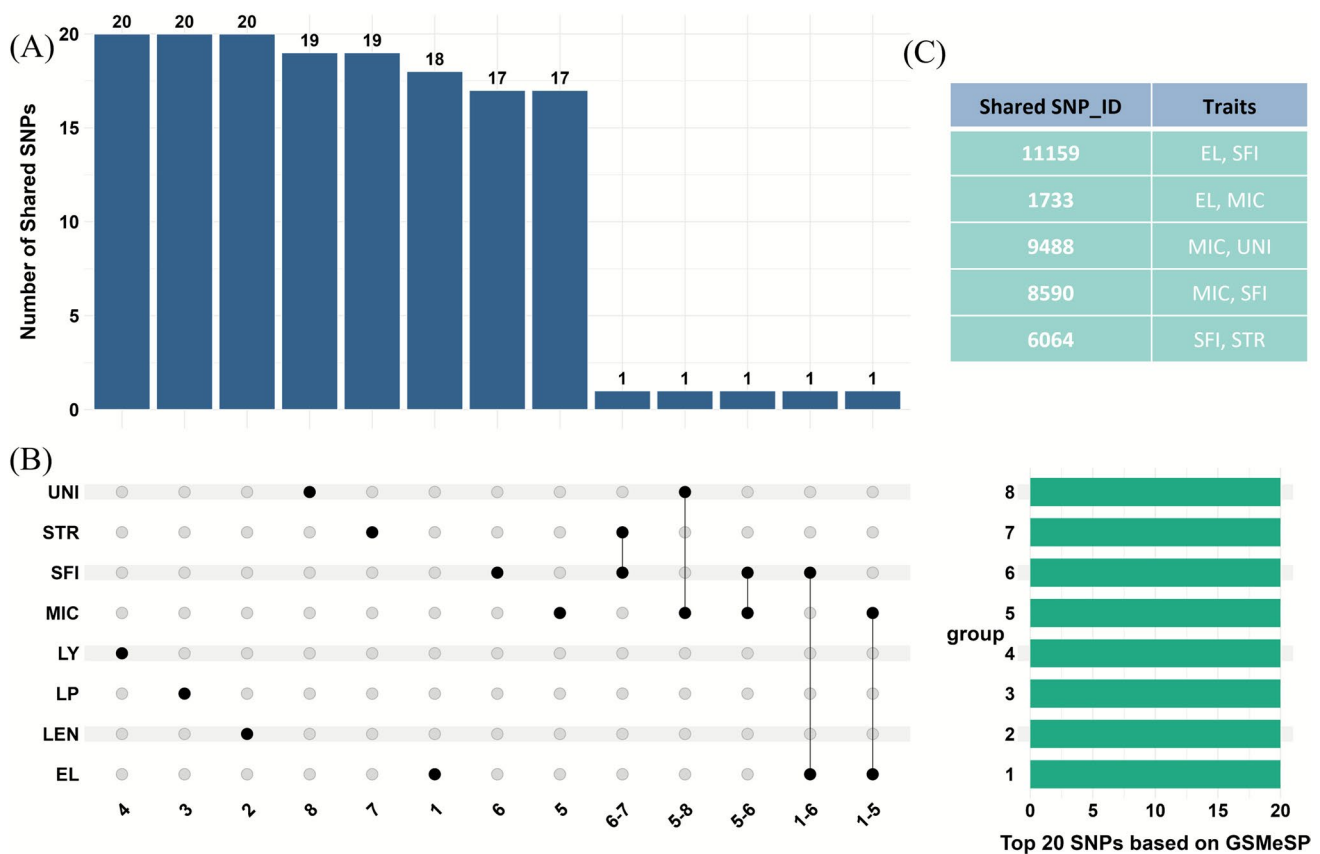


Fig. 7 UpSet plot visualizing shared top 20 ranked SNPs across traits identified by the GSMeSP framework. **A** Top bar chart shows the number of SNPs shared among different combinations of traits. The y-axis represents the number of shared SNPs, aligned with the x-axis categories in panel (B). **B** Matrix layout indicates which traits

are involved in each intersection: filled dots represent traits, and lines connect traits within each intersection. The y-axis lists all eight traits, and the right-side bar chart shows the total number of top-ranked SNPs per trait. **C** Accompanying table lists the specific SNP IDs in each intersection and the corresponding traits

SHAP quantifies the contribution of each SNP to model predictions, capturing both linear and nonlinear effects, as well as SNP–SNP interactions that traditional models may overlook (Ponce-Bobadilla et al. 2024). Our SHAP results revealed the top-ranked SNPs for each trait, providing a biologically meaningful prioritization of genomic features. These results offer actionable insights for breeders by highlighting SNPs with consistently large effects, while also revealing the directionality of allele influences, enabling the selection of favorable alleles and avoidance of deleterious variants (Rao et al. 2025; Yang et al. 2025). The contrasting patterns based on the distribution of SHAP values across traits revealed clear differences in genetic architecture and informed breeding strategy; for example, LP could be improved through targeted selection of key alleles, whereas STR improvement may require multi-trait selection indices or genomic selection to capture many small-effect loci.

Detecting pairwise gene interactions is a critical yet challenging aspect of dissecting complex traits, particularly when marginal effects are weak or uncorrelated with the phenotype (Fang et al. 2017). Our SHAP analysis revealed nonlinear SNP–SNP interactions that would remain undetected by conventional GWAS models based solely on marginal effects (Tam et al. 2019). For example, the interaction between SNPs 6351 and 12224 highlights SHAP's ability to uncover hidden layers of genetic architecture contributing to quantitative variation. SNP 12224 maps to Gh_D13G2343, a cotton homolog of *Arabidopsis thaliana* SOL1 (AT3G22760), which encodes a CXC domain-containing protein expressed in stamens, pollen mother cells, and ovules, where it regulates cell fate transitions, cell divisions, and floral organ development (Sijacic et al. 2011; Lu et al. 2013). The SHAP-identified interaction suggests that allelic variation at Gh_D13G2343 may conditionally modulate the influence of other loci on developmental pathways,

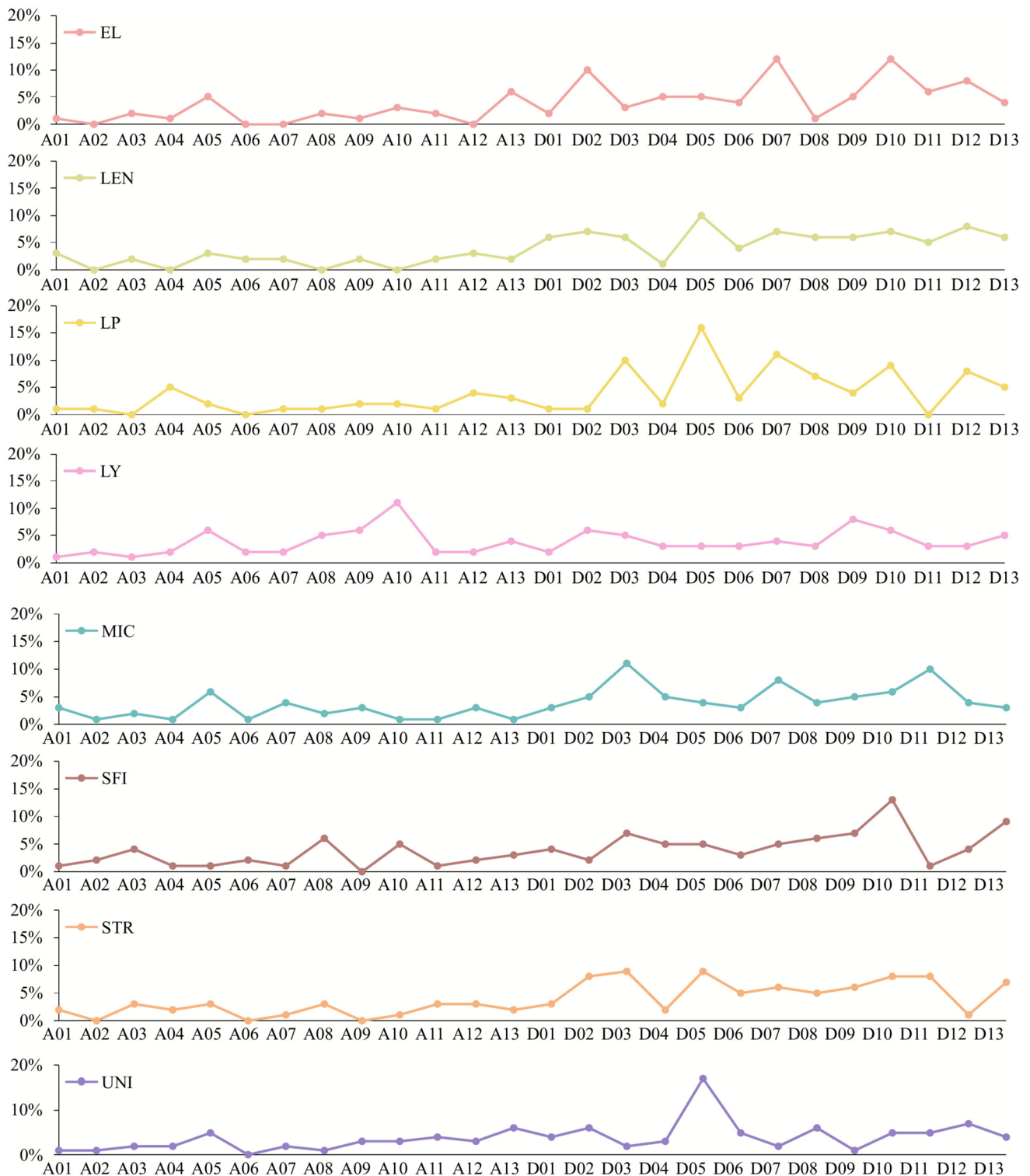


Fig. 8 Chromosome-wise carrier rate of the top 100 SNPs per trait based on the GSMeSP framework. The x-axis indicates chromosome numbers, while the y-axis shows the chromosome carrier rate. Different colored lines represent individual traits

thereby linking floral differentiation processes to fiber initiation and elongation. This positions SNP 12224 as a potential

regulatory node with both predictive and mechanistic relevance for cotton fiber traits.

To further enhance biological validity, we integrated SHAP outputs with GWAS results to generate a composite SNP ranking within GSMeSP, reflecting both predictive importance and statistical significance. While GWAS is powerful for detecting large-effect loci, it can overlook variants involved in complex interactions and is susceptible to confounding by population structure (Tam et al. 2019; Vellarino et al. 2023). This integrative prioritization balances interpretability with biological plausibility and highlights loci of high interest, either due to large individual effects or through involvement in complex interactions. SNPs with high SHAP scores and strong GWAS significance, such as MIC and SFI's shared SNP 8590 located near *GhCesA4-I-D07* (GH_D07G2083), a known fiber synthesis gene in cotton (Wang, et al., 2023c), represent strong candidates for functional validation. Conversely, SNPs with strong SHAP signals but modest GWAS significance may represent novel or underexplored regulators worthy of further investigation.

Chromosomal prioritization and trait architecture insights

In addition to SNP-level insights, GSMeSP supports chromosomal-level prioritization, guiding breeders toward key genomic regions. In our cotton dataset, chromosome D05 emerged as a genomic hotspot, contributing substantially to LEN, LP, and UNI traits, suggesting the presence of key regulatory loci involved in fiber development. Chromosomal distribution patterns varied across traits, reflecting their distinct genetic architectures. For instance, UNI appeared to be influenced by a smaller number of loci, primarily on D05, while LP was associated with broader contributions from several D-subgenome chromosomes. These observations are consistent with previous studies that highlight the enrichment of fiber-related genes on D05 and across the D-subgenome (Song et al. 2019; Ma et al. 2021). Importantly, chromosomal patterns derived from SHAP did not always align with those highlighted by GWAS alone, underscoring the complementary nature of both approaches. Integrating both enables a more complete understanding of trait regulation and ensures that both additive and interaction-driven effects are captured.

Future directions

This study demonstrated that BNNs deliver superior accuracy in multi-trait genomic prediction across a diverse set of cotton phenotypes. Complementing this, the GSMeSP framework offers a practical and interpretable approach for incorporating deep learning into genomic selection pipelines. However, further refinement is needed for both components to maximize their impact in breeding programs. The predictive performance of BNNs should be validated using

more diverse and representative datasets across multiple environments. In particular, incorporating G×E interactions through environmental similarity matrices or latent covariate models may significantly enhance model generalizability under variable field conditions (Montesinos-López et al. 2018). Future work on GSMeSP could focus on developing adaptive weighting strategies that adjust the contribution of SHAP and GWAS components based on trait complexity, GWAS signal strength, or biological priors. Additionally, linking GSMeSP-ranked SNPs with functional genomic databases or transcriptomic profiles would enable biological validation and improve the interpretability of model outputs. Enrichment analyses of gene ontologies and biological pathways could further uncover mechanisms underlying trait variation and guide genome editing or targeted breeding strategies (Zhou et al. 2024). Together, these advancements would enhance the biological relevance, transparency, and practical utility of deep learning-based genomic prediction, bridging the gap between statistical modeling and functional genomics, and ultimately supporting more informed and effective breeding decisions.

Conclusion

This study addresses two major limitations in the application of deep learning to plant breeding: the need for reliable uncertainty estimation and enhanced model interpretability. We proposed LASSO-regularized BNNs that achieved superior multi-trait genomic prediction accuracy across a diverse set of cotton phenotypes, with improvements ranging from 0.46 to 47.85% over traditional and deep learning-based methods in cotton dataset. The Bayesian framework enabled explicit modeling of predictive uncertainty, providing breeders with a robust means to assess the reliability of genotype-level predictions and to make risk-informed selection decisions. To complement this, the GSMeSP interpretability framework, which integrates SHAP-based feature attribution with GWAS evidence, provided biologically grounded insights into SNP importance, facilitating the identification of trait-associated markers and revealing complex genotype–phenotype relationships. Together, these contributions advance the development of deep learning models that are not only accurate but also transparent and actionable. As breeding programs increasingly adopt high-throughput genotyping and phenotyping technologies, integrated approaches such as the one presented here will be essential for guiding the selection of high-performing and resilient crop varieties in the face of evolving environmental and market pressures.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00122-025-05127-z>.

Acknowledgements The research was supported by the National Natural Science Foundation of China under Award No. 32401921, as well as from the Hebei Natural Science Foundation under Grant No. C2022203005.

Author contributions JS designed the overall methodology and drafted the manuscript; XZ and JS conducted the BNNs, SHAP, and GWAS analyses; XY and JS performed the LASSO and DNNGP analysis; and JC, MES, OM, and AM provided analysis suggestions and assisted with manuscript revisions.

Funding This study was funded by the National Natural Science Foundation of China, No. 32401921, and the Hebei Natural Science Foundation, No. C2022203005.

Data availability The datasets analyzed in this study are publicly available from the cited references. The code implementing the Bayesian neural networks (BNNs) framework, including uncertainty quantification and SHAP analysis, is openly accessible on GitHub at <https://github.com/GSBreeder/BNNs/tree/main>.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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